Q P CODE: 25900651

Scheme for Valuation

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCSTA203 - STATISTICAL RESEARCH TECHNIQUES USING SOFTWARES

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

PART A. Short answer type questions
Answer any 7 questions. Each question carries 2 marks

1. Definition /Statement/Use — 2 mark
2. Any two applications — 1 mark each
3. Definition/ Example — 2 mark
4. Definition — 1 mark each
5. Definition/ Example — 2 mark
6. Any two assumptions in Z-test — 1 mark each
7. Any one advantage and one disadvantage — 1 mark each
8. Any two assumptions of t test (S.D unknown, n small,normality etc) — 1 mark each
9. Any two important applications of t distribution in small sample test — 1 mark each
   (testing mean, testing equality of means in 2 populations etc)
10. Definition of null and alternate Hypothesis in ANOVA — 1 mark each

Part B (Short Essay Type Questions)
Answer any 4 questions. Each question carries 6 marks

11. Definition - 4 marks;
    Any one advantage - I mark; one disadvantage – 1 mark

12. Explanation of exploratory research design - 4 marks ; Example – 2 mark

13. Statement/ Definition of critical region- 2 marks
    . Explanation/ Demonstration of critical regions of one tailed test - 2 mark
    Explanation/ Demonstration of critical regions of two tailed test -2 mark

14. Each step of statistical test – 1.5 marks (Maximum marks limited to 6).

15. Null Hypothesis- **1 mark** (There is no significant difference in mean IQ before and after training (training not effective).)

Identification and writing of correct test statistic -Paired t Test $t = \frac{\overline{U}}{S_u\sqrt{n-1}}$ follows students t with n-1 d.f - **(1 marks)**

Mean U - 2 Variance of U = 24 ; t = 0.8164 **(2 mark)**

at 5% level (two-tailed),tabled value=2.776. **(1 mark)**
No reason to Reject null hypothesis - conclusion . - **(1 mark)**

16. Null Hypothesis- **1 mark** (Performance in exam is independent of marital status)

correct test statistic $\chi 2 = \sum \frac{(O_i - E_i)^2}{E_i}$ follows $\chi 2$ with (r-1)(c-1) d.f - **(1 marks)**

$\chi 2$ = 0.286+0.190+0.667+0.444=1.587 **(2 mark)**

at 5% level tabled value=3.841 **(1 mark)**
No reason to Reject null hypothesis - conclusion . - **(1 mark)**

Part C

Answer any 1 question (Each question carries 12 marks)

17. Any 6 conditions/ suggestions **2 mark each**

18. **Either sign test or Wilcoxon signed rank test**

A) Sign Test

Compute the difference for each student:

5, 4, −1, 1, 4, 1, 5, 2, 3, −13, −8, 11, −7, −9, 1, 2, −1, 2, 9, 3, 2, −2, −3, 1, 5, −3, −4 **2 marks**

No. of positive differences **-17,** No.of negative differences : **10,** Number of ties: **0; n = 27**
- **4 marks**

test statistic- X- no.of positive values **2 marks**

H_0 : (no median change)

X∼Binomial(27,0.5)
Two-sided exact p-value = 0.2478. **3 marks**

Conclusion (α = 0.05): fail to reject H_0 . No statistically significant change. **1 mark**

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B) Wilcoxon signed-rank test (paired, uses magnitudes + signs)

- Absolute differences were ranked ---------- **5 marks**
- Sum of ranks for positive differences W+=221.5 **2mark**
- For n=27 the normal-approximation with continuity correction gives
  $z \approx 0.769$ and two-sided p-value ≈ 0.4420. 4 **mark**

Conclusion ($\alpha = 0.05$): fail to reject null hypothesis . No statistically significant change. **1 mark**